

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

```
clc;
clear;
close all;

% Butterworth IIR Low Pass Filter
Fs = 10000;
Fp = 1000;
Fst = 1500;

[N,Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
[b,a] = butter(N,Wn);

% Objective 2: Magnitude and Phase Response
figure;
freqz(b,a);
title('Magnitude and Phase Response');

% Objective 3: Impulse Response
figure;
impz(b,a,50);
title('Impulse Response');

% Step Response
figure;
stepz(b,a);
title('Step Response');

% Objective 4: Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```



